香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

ELEC 3310
Digital Fundamentals and System Design

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Outline Lecture 13 Finite State Machines, Further Examples

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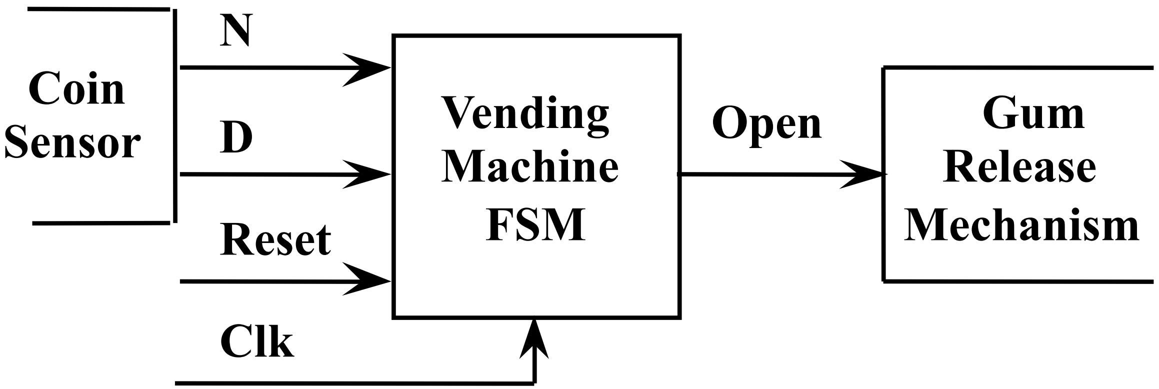
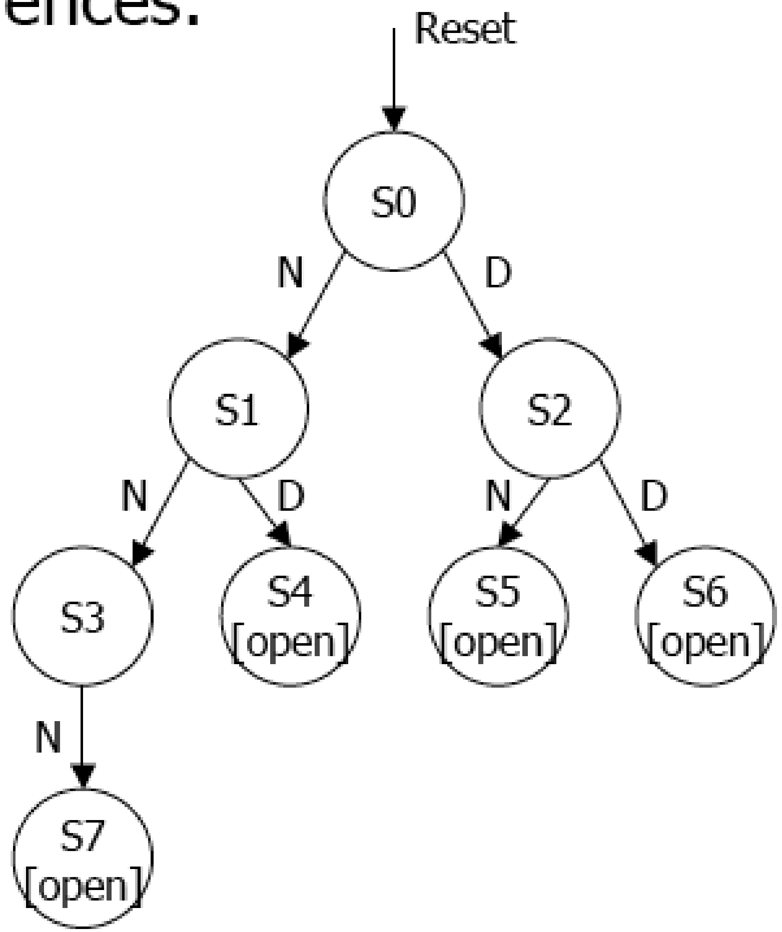
Vending machine

- **Project Specifications**
  - **Deliver a package of gum after 15 cents are deposited**
  - **Single coin slot for dimes (10 cents), nickels (5 cents)**
  - **Keep the change**
  - **More features are possible:**
    - » **out-of-gum condition, make change, accept quarters (25 cents) or dollars, choice of gum, ...**
- **Understand the Specifications**
  - **With a block diagram**
  - **2 data inputs**
  - **1 clock input**
  - **1 open output**
  - **1 reset input**
    - » **initial state**



Vending Machine - State Diagram

- Suitable Abstract Representation
  - Tabulate typical input sequences:
    - 3 nickels
    - nickel, dime
    - dime, nickel
    - two dimes
  - Draw state diagram:
    - Inputs: N, D, reset
    - Output: open chute
  - Assumptions:
    - Assume N and D asserted for one cycle
    - Each state has a self loop for N = D = 0 (no coin)



Vending Machine - State Diagram

Draw minimized state diagram

Vending Machine - State Reduction

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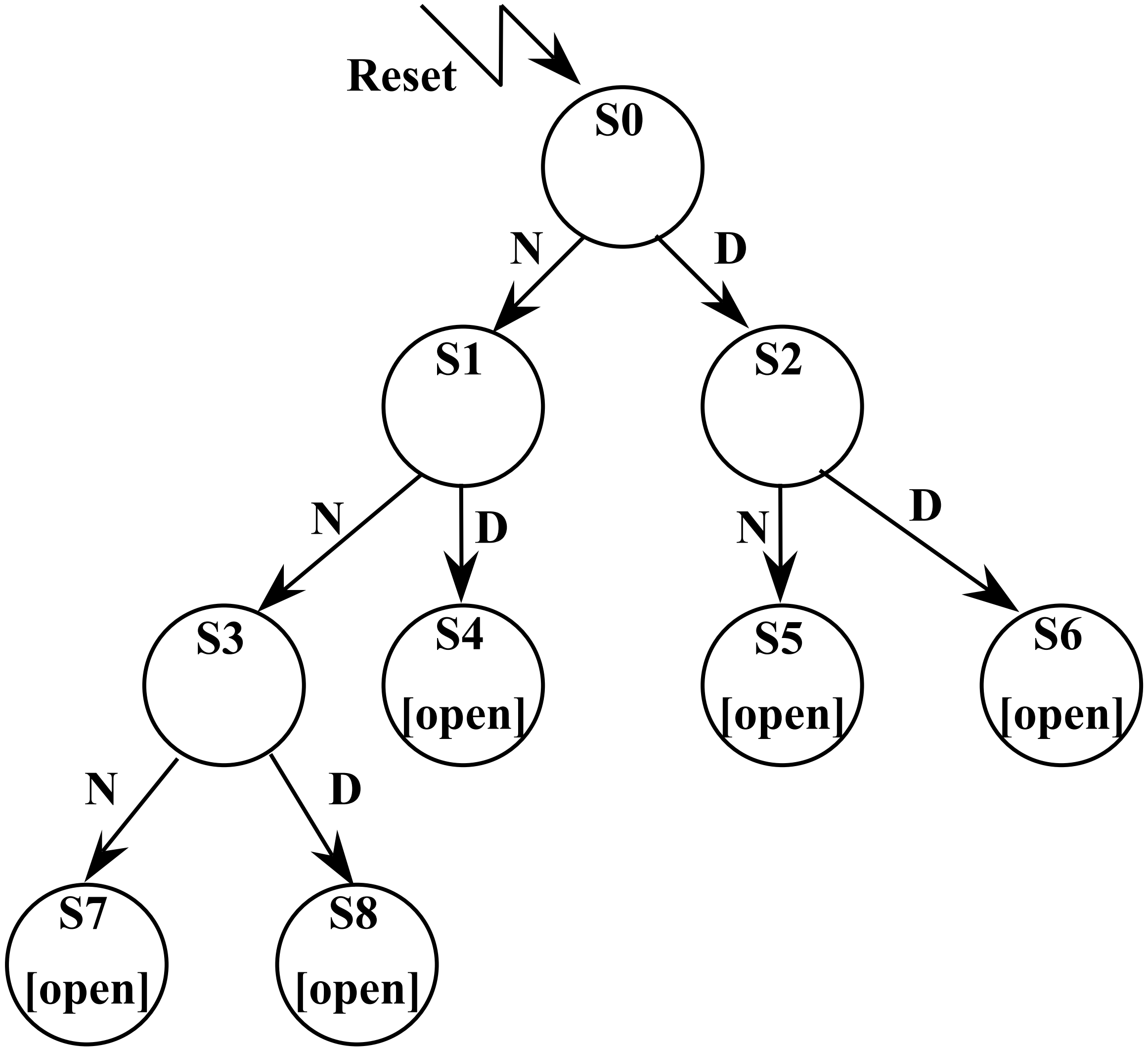
- Two FSM are equivalent if they produce the same output sequences given the same input sequences.
- The internal 'states' are not relevant from the observer point of view
- Hence, for a given state machine we can try to find an equivalent state machine, with fewer internal states (i.e. simpler? fewer flip-flops? perhaps…)
- The principle of FSM state reduction is based on *identifying and merging* equivalent states.
- "Two internal states are said to be equivalent if for each input combination they give exactly the same output and send the circuit to either the same state or to an equivalent state"

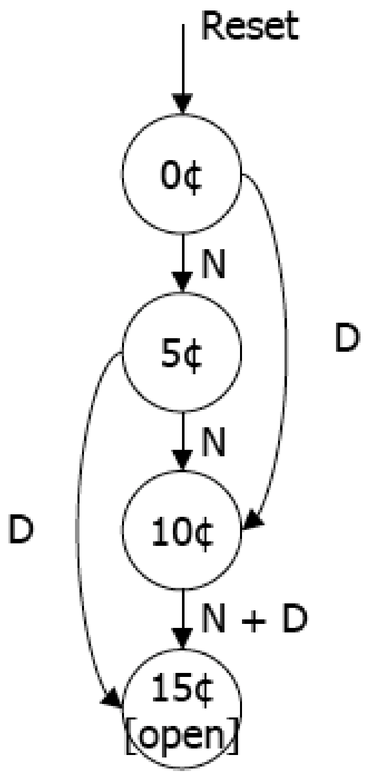
Vending Machine - State Reduction

- Perform state reduction
  - Equivalent states can be merged
    - » Same output
    - » Same state transitions
  - S4, S5, S6, S7, S8 are equivalent
    - » Same output [open]
    - » Same state transitions
      - reset to S0
  - S2 and S3 are equivalent
    - » Same output
    - » Same state transitions
  - 4 states
    - » S0 (0¢)
    - » S1 (5¢)
    - » S2’ (10¢)
    - » S4’ (15¢, open)



Vending Machine - State Reduction

- **Perform state reduction**
  - **Minimized state-transition diagram and table**
  - **(11) is not a possible input, don't care**
    - » **since two coins cannot be inserted at the same time**



present state	inputs D	inputs N	next state	output open
0¢	0	0	0¢	0
	0	1	5¢	0
	1	0	10¢	0
	1	1	–	–
5¢	0	0	5¢	0
	0	1	10¢	0
	1	0	15¢	0
	1	1	–	–
10¢	0	0	10¢	0
	0	1	15¢	0
	1	0	15¢	0
	1	1	–	–
15¢	–	–	15¢	1

symbolic state table

Vending Machine - State Assignment

- Perform state assignment
  - Assign binary values to the states
  - 4 states
    - » 2 flip-flops
  - State assignment
    - » 0 ¢ by 00
    - » 5 ¢ by 01
    - » 10 ¢ by 10
    - » 15 ¢ by 11
- Choose D flip-flop
  - D → Q^+

present state Q1 Q0	inputs D	N	next state D1	D0	output open
0 0	0	0	0	0	0
	0	1	0	1	0
	1	0	1	0	0
	1	1	–	–	–
0 1	0	0	0	1	0
	0	1	1	0	0
	1	0	1	1	0
	1	1	–	–	–
1 0	0	0	1	0	0
	0	1	1	1	0
	1	0	1	1	0
	1	1	–	–	–
1 1	0	0	1	1	1
	0	1	1	1	1
	1	0	1	1	1
	1	1	–	–	–

Vending Machine - Implementation

DN \ Q_1Q_0	00	01	11	10
00	0	0	1	1
01	0	1	1	1
11	X	X	X	X
10	1	1	1	1

K-map for D_1

$$D_1 = Q_1 + D + Q_0 N$$

DN \ Q_1Q_0	00	01	11	10
00	0	1	1	0
01	1	0	1	1
11	X	X	X	X
10	0	1	1	1

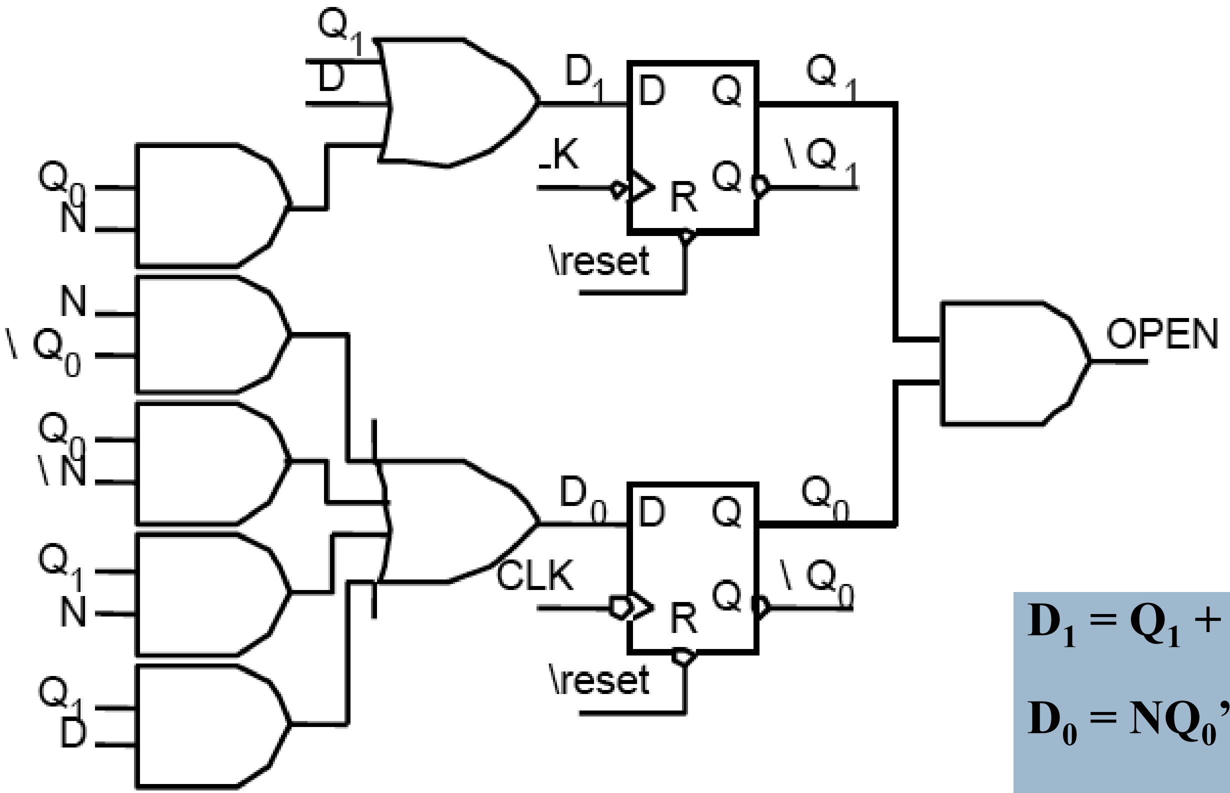
$$D_0 = NQ_0' + Q_0N' + Q_1N + Q_1D$$

DN \ Q_1Q_0	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	X	X	X	X
10	0	0	1	0

K-map for Open

$$OPEN = Q_1\, Q_0$$

Vending Machine - Implementation



$D_1 = Q_1 + D + Q_0\,N$

$D_0 = NQ_0' + Q_0N' + Q_1N + Q_1D$

$OPEN = Q_1\,Q_0$

Vending Machine – Implementation (JKFF)

Present State Q_1	Present State Q_0	Inputs D	Inputs N	Next State Q_1^+	Next State Q_0^+	J_1	K_1	J_0	K_0
0	0	0	0	0	0	0	X	0	X
		0	1	0	1	0	X	1	X
		1	0	1	0	1	X	0	X
		1	1	X	X	X	X	X	X
0	1	0	0	0	1	0	X	X	0
		0	1	1	0	1	X	X	1
		1	0	1	1	1	X	X	0
		1	1	X	X	X	X	X	X
1	0	0	0	1	0	X	0	0	X
		0	1	1	1	X	0	1	X
		1	0	1	1	X	0	1	X
		1	1	X	X	X	X	X	X
1	1	0	0	1	1	X	0	X	0
		0	1	1	1	X	0	X	0
		1	0	1	1	X	0	X	0
		1	1	X	X	X	X	X	X

Q	Q+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Vending Machine – Implementation (JKFF)

Present State Q_1	Q_0	Inputs D	N	Next State Q_1^+	Q_0^+	J_1	K_1	J_0	K_0
0	0	0	0	0	0	0	X	0	X
		0	1	0	1	0	X	1	X
		1	0	1	0	1	X	0	X
		1	1	X	X	X	X	X	X
0	1	0	0	0	1	0	X	X	0
		0	1	1	0	1	X	X	1
		1	0	1	1	1	X	X	0
		1	1	X	X	X	X	X	X
1	0	0	0	1	0	X	0	0	X
		0	1	1	1	X	0	1	X
		1	0	1	1	X	0	1	X
		1	1	X	X	X	X	X	X
1	1	0	0	1	1	X	0	X	0
		0	1	1	1	X	0	X	0
		1	0	1	1	X	0	X	0
		1	1	X	X	X	X	X	X

K-map for J_1

DN \ Q_1Q_0	00	01	11	10
00	0	0	X	X
01	0	1	X	X
11	X	X	X	X
10	1	1	X	X

$$J_1 = D + Q_0 N$$

K-map for K_1

DN \ Q_1Q_0	00	01	11	10
00	X	X	0	0
01	X	X	0	0
11	X	X	X	X
10	X	X	0	0

K1 = 0

Vending Machine – Implementation (JKFF)

Present State Q_1	Q_0	Inputs D	N	Next State Q_1^+	Q_0^+	J_1	K_1	J_0	K_0
0	0	0	0	0	0	0	X	0	X
		0	1	0	1	0	X	1	X
		1	0	1	0	1	X	0	X
		1	1	X	X	X	X	X	X
0	1	0	0	0	1	0	X	X	0
		0	1	1	0	1	X	X	1
		1	0	1	1	1	X	X	0
		1	1	X	X	X	X	X	X
1	0	0	0	1	0	X	0	0	X
		0	1	1	1	X	0	1	X
		1	0	1	1	X	0	1	X
		1	1	X	X	X	X	X	X
1	1	0	0	1	1	X	0	X	0
		0	1	1	1	X	0	X	0
		1	0	1	1	X	0	X	0
		1	1	X	X	X	X	X	X

K-map for J_0

DN \ Q_1Q_0	00	01	11	10
00	0	X	X	0
01	1	X	X	1
11	X	X	X	X
10	0	X	X	1

$$J_0 = N + Q_1 D$$

K-map for K_0

DN \ Q_1Q_0	00	01	11	10
00	X	0	0	X
01	X	1	0	X
11	X	X	X	X
10	X	0	0	X

$$K_0 = Q'_1 N$$

Vending Machine – Implementation (JKFF)

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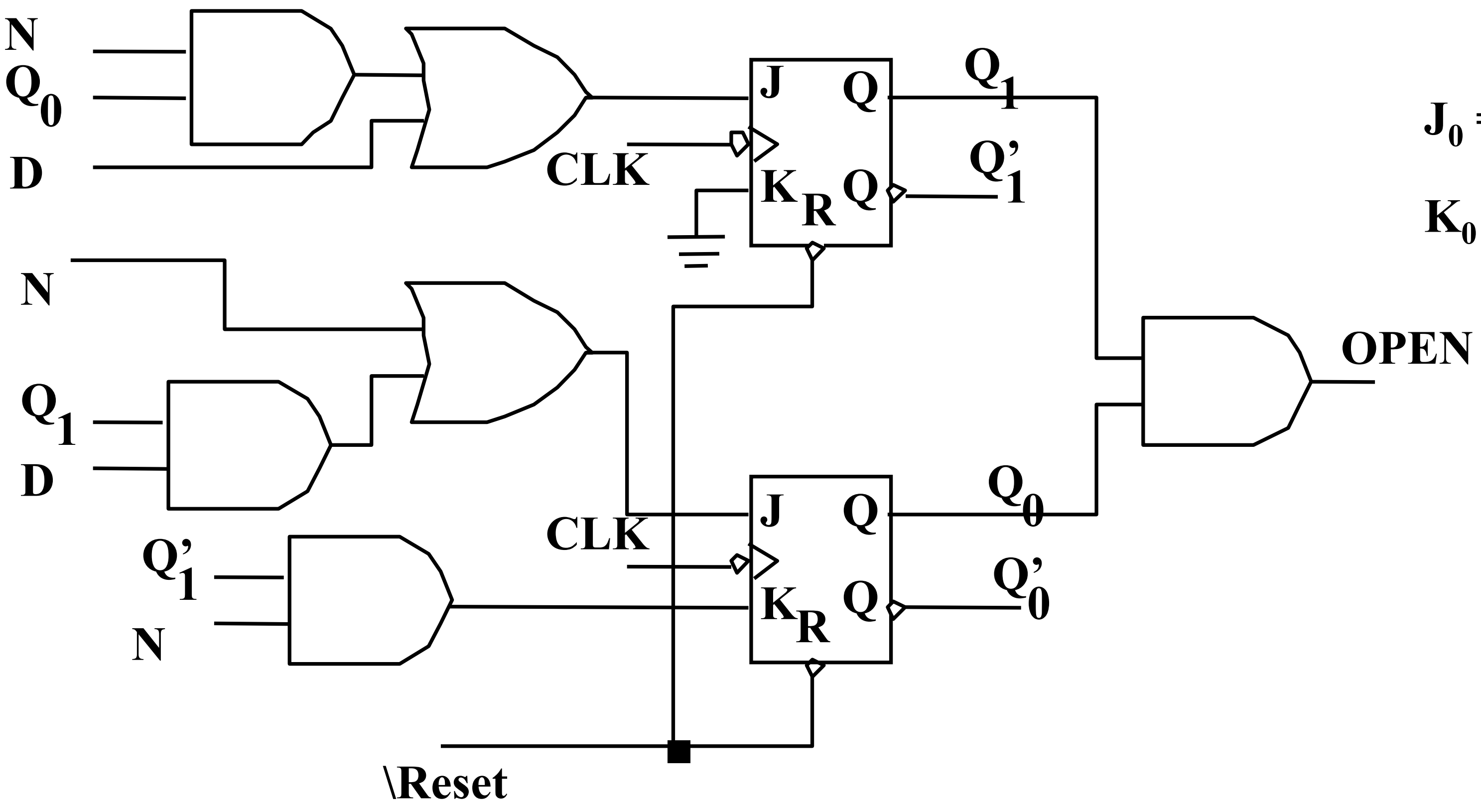


$J_1 = D + Q_0 N$

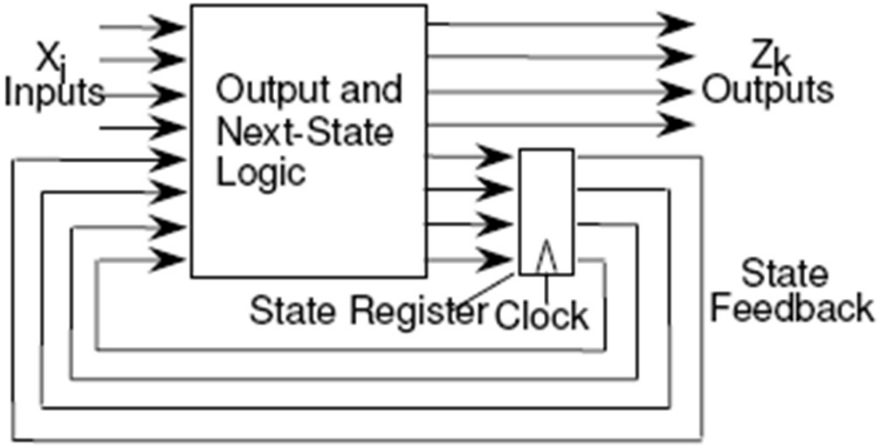
$K1 = 0$

$J_0 = N + Q_1 D$

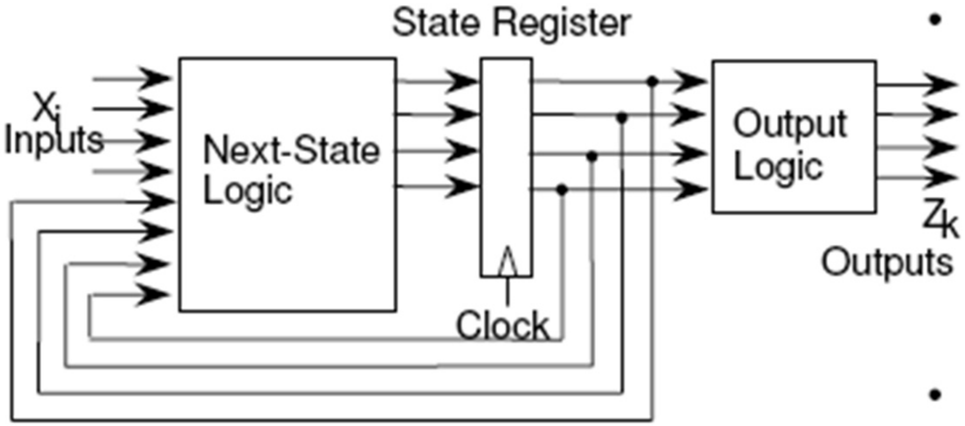
$K_0 = Q'_1 N$

Moore and Mealy Machines

- **Moore Machine**
  - **Outputs are function solely of the current state**
  - **Outputs change synchronously with state changes**
- **Mealy Machine**
  - **Outputs depend on state AND inputs**
  - **Input change causes an immediate output change**



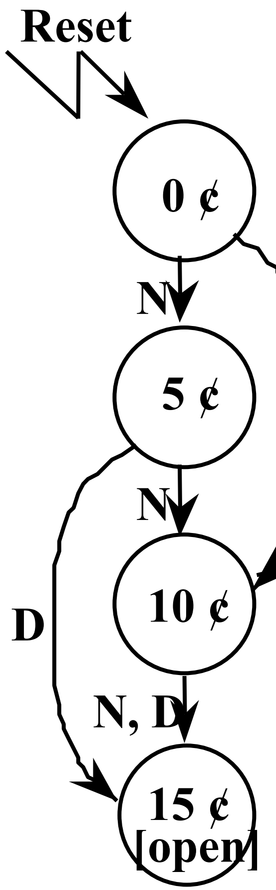
Mealy Machine

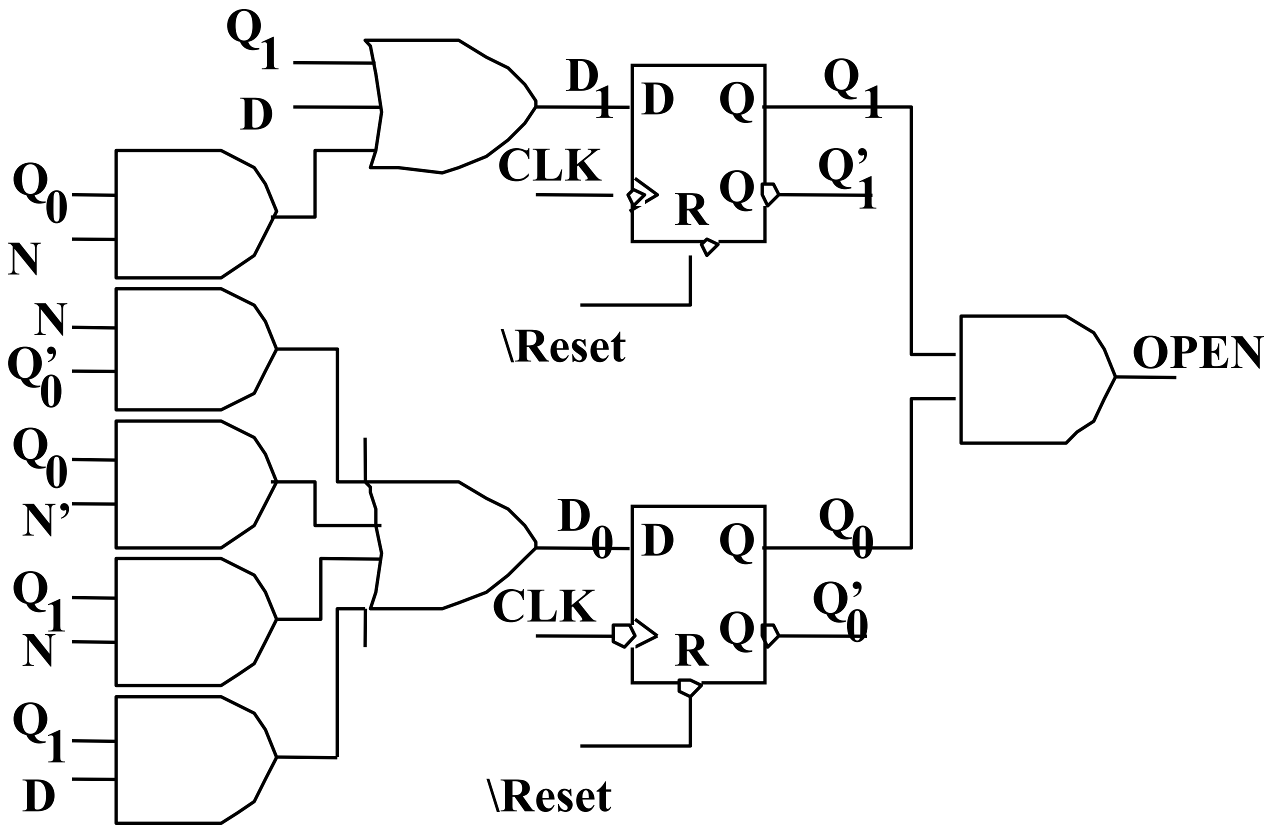


Moore Machine

The Example Vending Machine is a Moore Machine

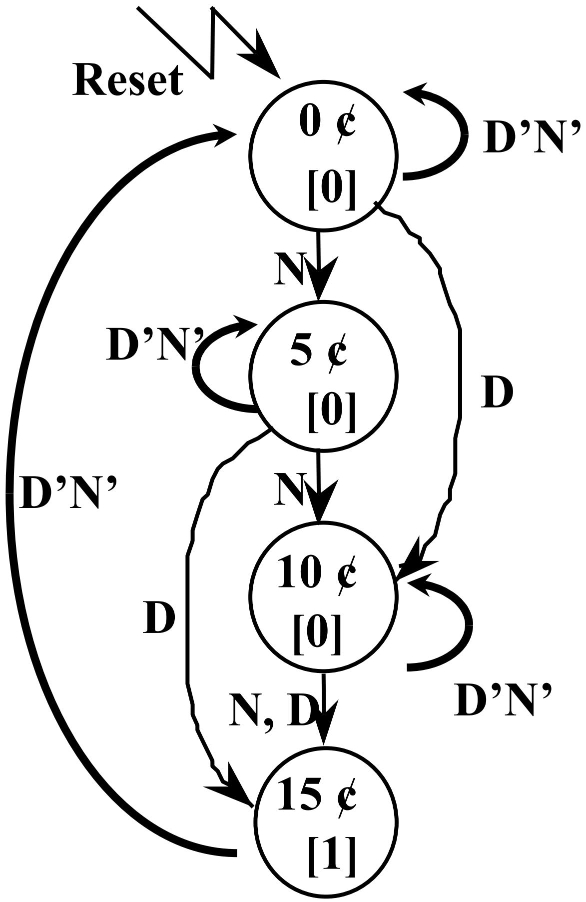
- Output is a function of the current states
  - output is "in circle" in the state transition diagram
  - Outputs change synchronously with states





A Better Expression

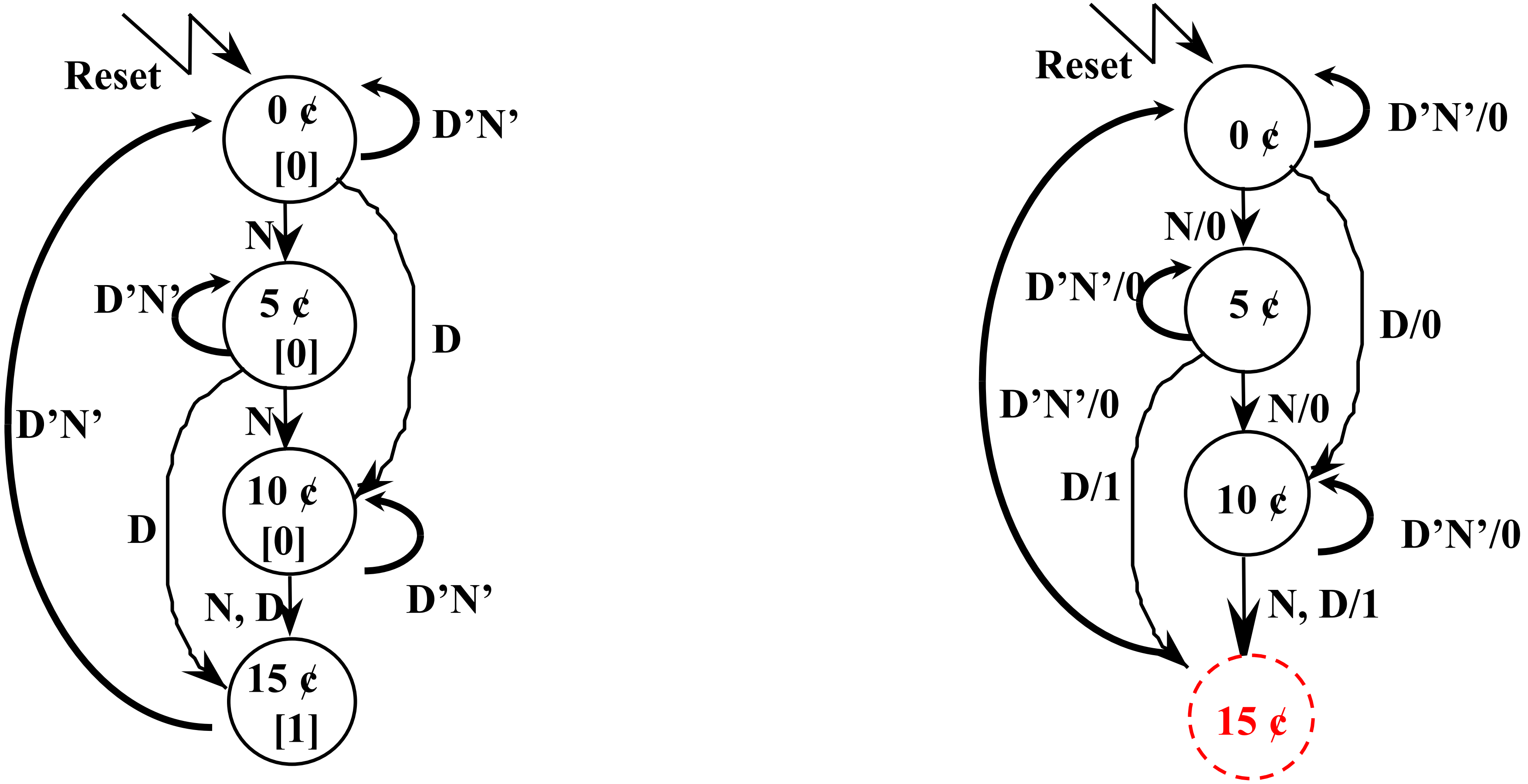
– **Output is associated with current state in every circle**
– **After gum is released, 15 cents shall be transited to 0 cents**



Present State	Inputs D	Inputs N	Next State	Output Open
0 ¢	0	0	0 ¢	0
	0	1	5 ¢	0
	1	0	10 ¢	0
	1	1	X	X
5 ¢	0	0	5 ¢	0
	0	1	10 ¢	0
	1	0	15 ¢	0
	1	1	X	X
10 ¢	0	0	10 ¢	0
	0	1	15 ¢	0
	1	0	15 ¢	0
	1	1	X	X
15 ¢	0	0	0 ¢	1
	0	1	X	1
	1	0	X	1
	1	1	X	1

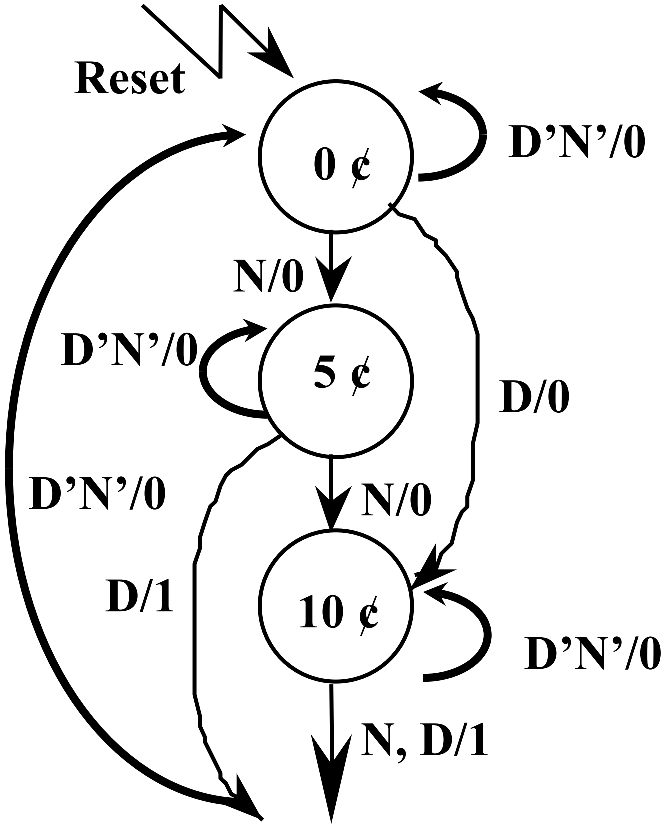
Transfer to the Mealy Machine

- Transfer to Mealy machine by moving output to the branches
- Mealy machine tends to have fewer states
  - » Do we need state “15 cents”?



A Mealy Machine

- It really has fewer states that the Moore machine
  - » Use Mealy machine if it is not specified
  - » State “15 cents” can be combined with state “0 cents”
- Also note the output difference



Present State	Inputs D	Inputs N	Next State	Output Open
0 ¢	0	0	0 ¢	0
	0	1	5 ¢	0
	1	0	10 ¢	0
	1	1	X	X
5 ¢	0	0	5 ¢	0
	0	1	10 ¢	0
	1	0	0 ¢	1
	1	1	X	X
10 ¢	0	0	10 ¢	0
	0	1	0 ¢	1
	1	0	0 ¢	1
	1	1	X	X

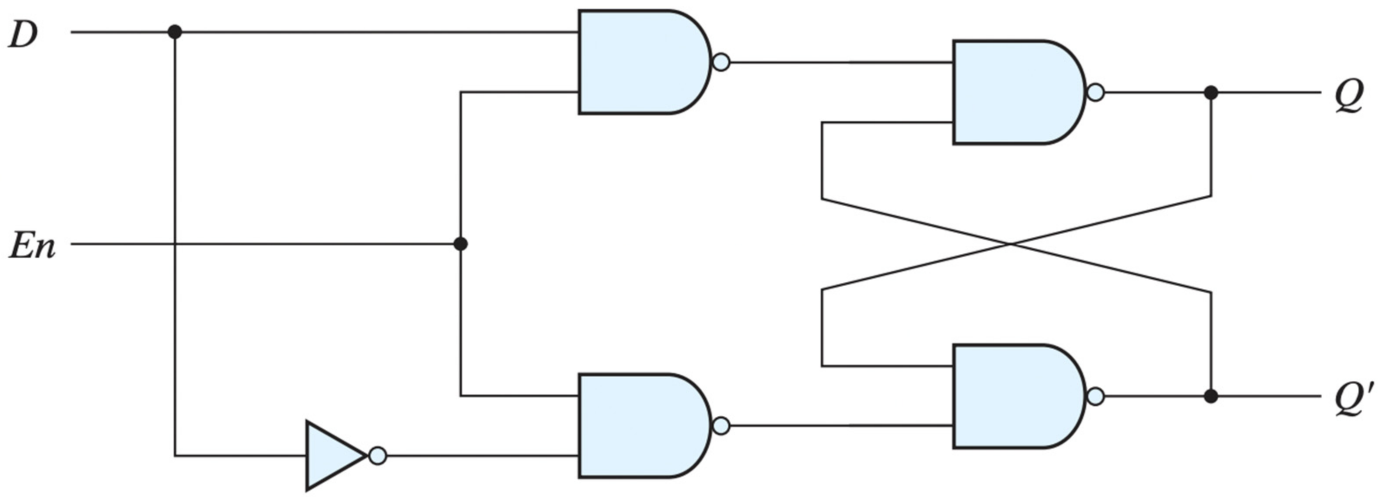
Latch

The D latch below is constructed with four NAND gates and an inverter. Consider the following three other ways for obtaining a D latch. In each case, draw the logic diagram and verify the circuit operation.

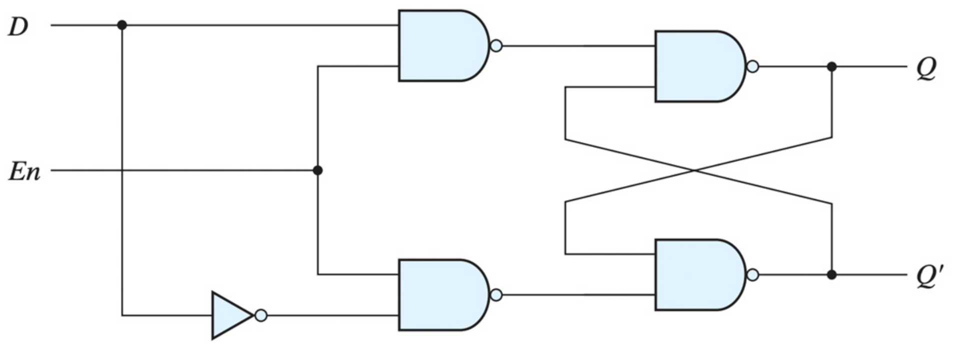
(a) Use NOR gates for the SR latch part and AND gates for the other two. An inverter may be needed.

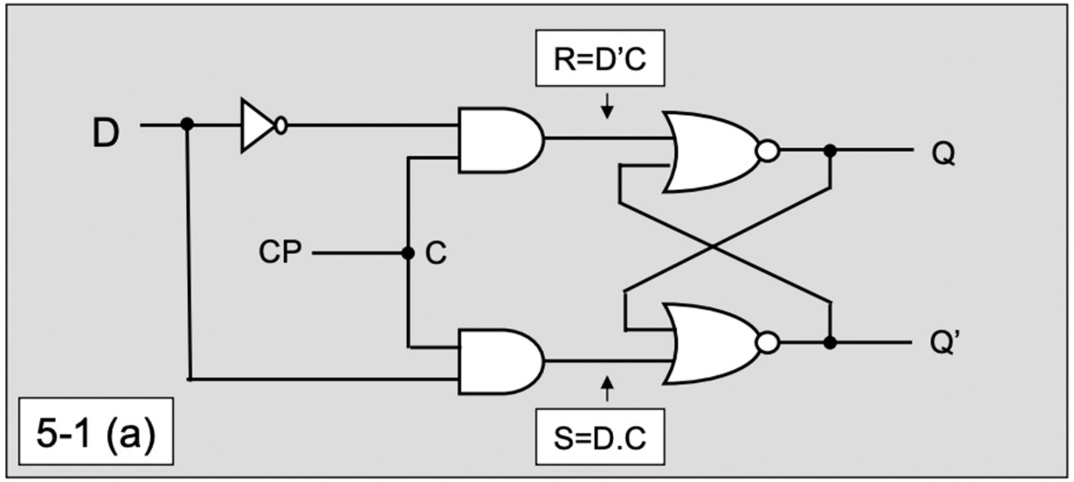
(b) Use NOR gates for all four gates. Inverters may be needed.

(c) Use four NAND gates only (without an inverter). This can be done by connecting the output of the upper gate in below figure (the gate that goes to the SR latch) to the input of the lower gate (instead of the inverter output).

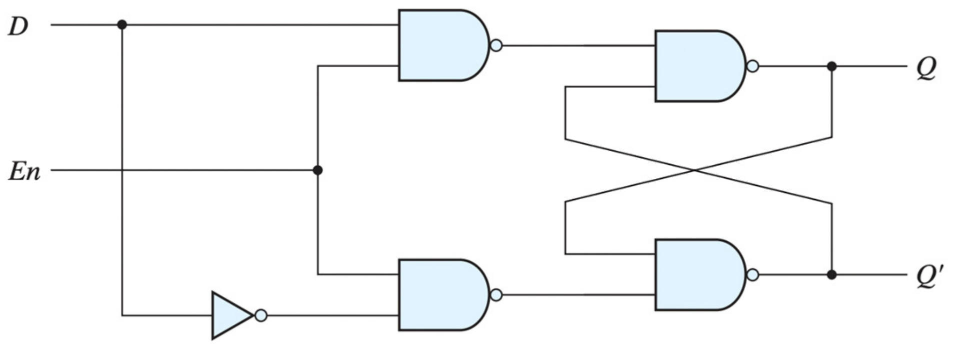


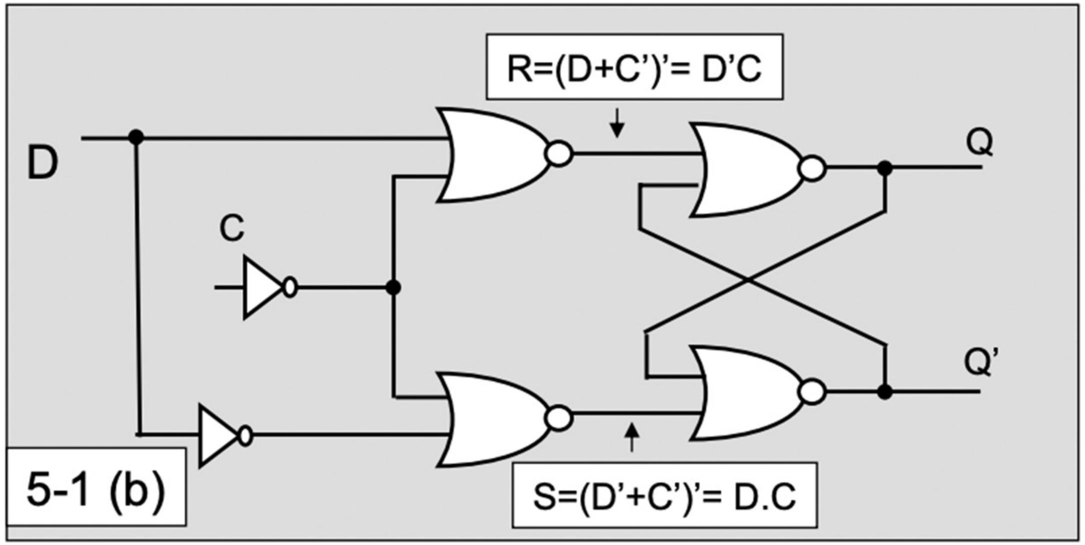
Latch



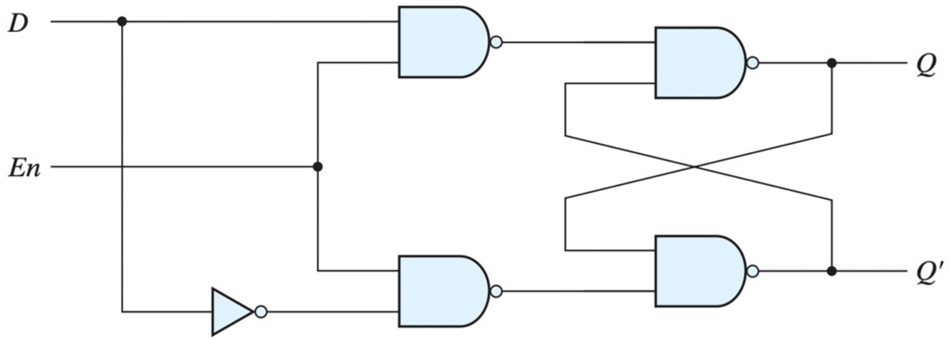


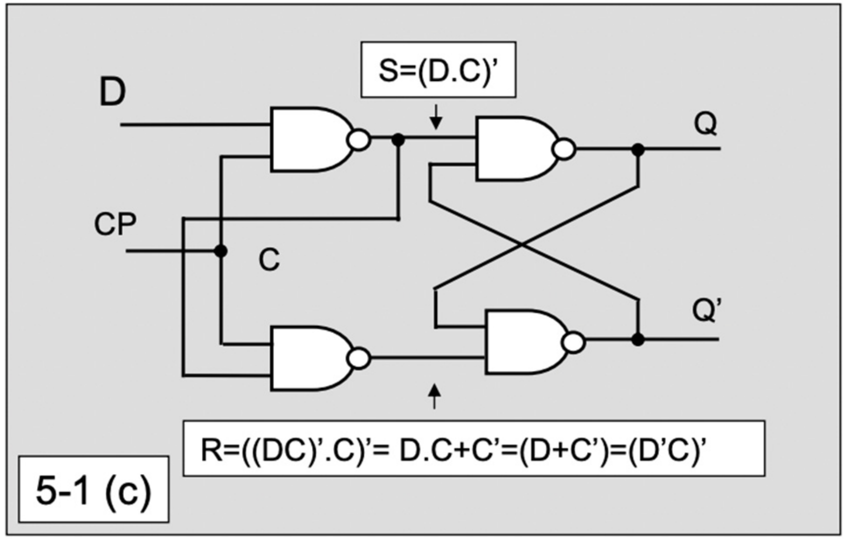
Latch





Latch





A PN flip-flop has four operations: clear to 0, no change, complement, and set to 1, when inputs P and N are 00, 01, 10, and 11, respectively.

(a) Tabulate the characteristic table.
(b) Derive the characteristic equation.
(c) Tabulate the excitation table.
(d) Show how the PN flip-flop can be converted to a D flip-flop.

FF

P	N	Q(t+1)
0	0	0
0	1	Q(t)
1	0	Q'(t)
1	1	1

5-4(a)

P	N	Q(t)	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

5-4(b)

Q(t)	Q(t+1)	P	N
0	0	0	X
0	1	1	X
1	0	X	0
1	1	X	1

5-4(c)

5-4(d)

Connect P and N together

Q(t+1)= PQ'+NQ

			N	N
			1	
P	1		1	1
		Q(t)	Q(t)	

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Thank you for listening !